

File Passwords

For several common file formats, there are tools available to extract forgotten passwords. Elcomsoft (www.elcomsoft.com) probably has the longest portfolio of such software, and their tools have proven to work in most cases. They have tools to recover passwords from Word, Excel, PowerPoint, Access, Outlook, Zip, RAR and many more file formats. All one needs to do is run the software, set a few password criteria, choose the file and let the software run. Using long and complicated passwords would make it quite difficult for the software to crack the password.

Archives such as Zip and RAR files have a very weird behaviour. If you set a pass-

word when creating the archive, the password is assigned to each file in the archive and not the archive itself. Thus, the file can be opened and the entire list of files can be seen. The files cannot be extracted without the password, but in many cases, viewing the file listing itself could be detrimental. One way to make them more secure is to compress files into an archive with a password, and then compress that archive into another one, again with a password. Thus, the only file that can be seen will be the original archive. Its contents will not be visible until a password is provided to extract it.



networked workstation. The first traces of network protocols, clients and services were thrown into Windows for Workgroups 3.11. Windows 95 had a much better network implementation, but the protocols used have gone through quite a few changes. Primitive protocols have been dropped and a handful of protocols such as TCP/IP and IPX/SPX have come up as winners by natural selection.

NetBIOS (Network Basic Input/Output

usually the first port hackers try to access—password-protecting your shares only make it slightly more difficult to get to your files. There are ample tools on the Net such as GFI LANguard Network Security Scanner (www.gfi.com) that can discover passwords in a matter of seconds.

Your computer is vulnerable even if you are not on a LAN and don't have any shared folders or printers, as long as the service is installed. The best solution would be to remove the File and Print Sharing service, if you are not going to be using these services. To do this, open **Control Panel > Network**, click File and Print Sharing and disable both the checkboxes. If Client for Microsoft Networks is installed, you will have to remove that too. You can then select TCP/IP and click Properties and check the NetBIOS tab to ensure that it has been disabled. Check the same for other protocols such as IPX/SPX and remove the NetBEUI protocol, which is just an extension of NetBIOS. However, you cannot do this if you are on a LAN and need to share resources. There is no real solution to prevent discovery of passwords on a LAN. User-level access is slightly stronger than share-level access, especially when logging in to a Windows NT or 2000 domain. Using long and complicated passwords also makes them difficult to figure out. However, if you want reliable security for your files, you should use some file encryption tools such as PGP (www.pgpi.org). (For more on encryption, data security and PGP, refer to *Datawatch in the November 2002 Digit issue*).

Content Advisor

The parental control features added to Internet Explorer are a boon in many

households and organisations. Access to Web sites using Internet Explorer can be restricted to a great extent, based on various settings. In Internet Explorer 4 however, there were several bugs in this system that caused problems with the Content Advisor settings. The feature would enable or disable itself, forget the password and manifest other problems.

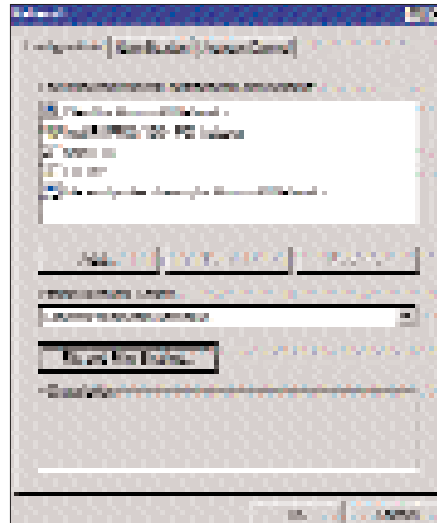
With each new problem that surfaced, Microsoft had to put up documentation on how to solve it. Most of the solutions gave detailed step-by-step instructions on how to disable the Content Advisor and reset its password, without requiring the current password. One such document at <http://support.microsoft.com/default.aspx?scid=kb;en-us;242037> lists virtually every version of Internet Explorer to be affected by the bug. These solutions also found their way to various support sites and message boards. Thus, it became quite simple to bypass this security.

In this case, the security feature is not



Snitch monitors Web activity for adult content in real time

inherently flawed, but ways to circumvent it are well documented. Thus, there is no workaround within Internet Explorer that will strengthen the security. If you want to restrict access to certain Web sites, you will have to use a third-party parental control software. Cyber Patrol (www.surfcontrol.com) filters out indecent or otherwise unwanted material on the Internet by controlling the user's Internet access and can also restrict the amount of time spent online. It updates the list of inappropriate sites daily and provides some powerful filtering options, giving you adequate control over each Internet session. Snitch (www.hyperdynesoftware.com), another Web-control software, works on a very interesting concept. It filters out content in real-time, by using several algorithms to detect adult content. It looks for words of a certain kind and analyses images to determine what per cent of them is showing skin, etc. It can even



Configure network services and protocols

System) has been adopted as an industry standard to access LAN resources. Windows 98 uses a very primitive protocol, LAN Manager, which is quite weak on security. Microsoft implements NetBIOS over TCP/IP to share resources. The File and Print Sharing service primarily runs on port 139 and can easily give access to anyone on the network—LAN, or even the Internet. This is